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PEEKABOO! THE EFFECT OF DIFFERENT VISIBLE CASH DISPLAY AND AMOUNT OPTIONS DURING MAIL CONTACT WHEN RECRUITING TO A PROBABILITY-BASED PANEL

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Recent studies consistently showed that making cash visible with a windowed envelope during mail contact increases response rates in surveys. The visible cash aims to pique interest and encourage sampled households to open the envelope. This article extends prior research by examining the effect of additional interventions implemented during mail recruitment to a survey panel on recruitment rates and costs. Specifically, we implemented randomized experiments to examine size (small, large) and location (none, front, back) of the window displaying cash, combined with what part of the cash is shown through the window envelope (numeric amount, face/image), and various prepaid incentive amounts (two $1, one $2, one $5). We used the recruitment effort for NORC's AmeriSpeak Panel as the data source for this study. The probability-based AmeriSpeak Panel uses an address-based sample and multiple modes of respondent contact, including mail, phone, and in-person outreach during recruitment. Our results were consistent with prior research and showed significant improvement in recruitment rates when cash was displayed through a window during mail contact. We also found that placing the window on the front of the envelope, showing

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$5 through the envelope compared to $2 and $1, and showing the tender amount compared to the image on the cash through the window were more likely to improve the recruitment rates. Our cost analyses illustrated that the cost difference in printing window versus no window envelope is small. There is no difference in printing cost between front window and back window as they both require custom manufacturing. There is also no cost difference in printing envelopes with small windows versus large windows. Lastly, we found no evidence of mail theft based on our review of the United States Postal Service's "track and trace" reports, seed mailings sent to staff, and undeliverable mailing rates.

KEY WORDS: Prepaid cash incentive; Probability-based panel; Recruitment mailing; Recruitment rate; Visible cash.

Statement of Significance

Inclusion of cash incentives in the mailed survey requests is a common practice to increase survey response rates. Accordingly, researchers have tried innovative ways to make sure that the envelope is opened and cash in the envelope is not discarded. Based on recent empirical studies, the use of window envelopes that enable the cash to be visible through the envelope has been shown to be effective in increasing response rates. However, to our knowledge, prior research has not tested different designs of the envelope window or parts of the visible cash. We extend this research and explore four ways to vary the presentation of cash: cash amount, location of the window on the envelope, window size, and what part of cash is shown through the window. The experiment results illustrate that $5 shown through the envelope compared to $2 and showing the tender amount/number compared to the face/image on the cash through the window were more effective in terms of improving recruitment rates. Additionally, we have not found any evidence of mail theft based on our review of the "track and trace" reports and undeliverable rates.

1. INTRODUCTION

As survey research increasingly has moved toward a mixed-mode design approach (De Leeuw 2005, 2018; Dillman et al. 2014), whether potential respondents are "attentive" to the mailed survey request is a paramount concern. For address-based probability samples, participation becomes a three-step process including first, whether or not an adult within the sampled household opens the mailed envelope; second, whether they then consider the contents and message contained therein, which typically includes an invitation to participate in a web survey or via an included paper-and-pencil instrument; and finally, whether they participate. As such, researchers have long noted the importance of mailing material design, investigating methods to design the mailing envelope to stand out from junk mail, cue a recipient that it is worth opening, and encourage the sampled household to pay attention to the included materials (Dillman 1978; Heberlein and Baumgartner 1978; Dillman et al. 2014). Various envelope design strategies have been shown to impact respondents' decision to open the envelope and participate in surveys (Heberlein and Baumgartner 1978; Meredith 2004; Amos and Paswan 2009; Kolyesnikova et al. 2011; Dykema et al. 2015; Endres et al. 2023).

A recent experiment (DeBell et al. 2020) found a "visible cash window" where a \$5 cash incentive was shown through an envelope window improved survey response rates compared to a nonwindowed envelope. While they were successful in improving their response rate by 4.3 percentage points—and found no difference when the letter was sent via first class or priority mail—the study did not go beyond exploring the general overarching effect of visible cash shown through the envelope. A follow-up study (DeBell 2022) found showing a larger (\$10) incentive through the window envelope improved survey screener response rate by 6.7 percentage points and survey response rate by 4.8 percentage points compared to a nonwindowed envelope. The same study also examined the effect of displaying cash through the window envelope as part of a panel study, finding that the visible cash did not help response rates among the prior study participants. A third study found the use of window envelopes with \$2 visible cash during mail recruitment significantly increased overall response rate by 1.6 percentage points (Williams et al. 2022). Finally, Sherr and Wells (2021) found a slight (0.5 percentage point, $p = .06$) increase in their cooperation rates when \$2 visible cash incentive was utilized during mail recruitment for the California Health Interview Survey. Still, these studies also did not experiment with additional mail design interventions and how such interventions might interact with the "visible cash" method. Additionally, the varied strength across these experiments may indicate that different amounts of cash shown through the envelope affect completion rates, but these designs do not provide a true experimental comparison.

Much remains to be understood regarding the use of visible cash through windowed envelopes and its effect on survey and panel participation. There are many ways a prepaid cash incentive can be shown through the envelope to a potential respondent. To our knowledge, there are no studies that experimented with the effect of differential dollar amounts displayed through the window envelope; the size and location of the window on the envelope; or regarding the various ways cash can be displayed through the window, all of which may affect response rates. Thus, we explore four ways to vary the presentation of cash: (1) cash amount (two \$1 versus one \$2 versus one \$5); (2) location of the window on the envelope (none versus front versus back);

(3) window size (small versus large); and (4) what part of the cash is shown through the window (numeric amount versus face/image on the cash). Also, there is little known about whether visible incentives are effective during panel recruitment because panel recruitment differs in its request from a cross-sectional survey. This article will make these contributions to our understanding of the use of visible cash incentives on panel recruitment rates and costs.

1.1 Background and Hypotheses

Best practices for the use of visible cash incentives during survey participation requests are still in development. Prepaid cash incentives increase response to surveys [see systematic reviews of studies on survey incentives conducted by Singer and Ye (2013) and Mercer et al. (2015)]. Presumably, the more likely receivers know they are being given a prepaid cash incentive, the more effective those incentives can be. Leverage-salience theory (Groves et al. 2000) is useful to understand the effectiveness of incentives in surveys, and as such an important question to consider is how to increase the saliency of the incentive (by displaying it through a windowed envelope) to allow that incentive to have positive leverage (Groves et al. 2000). Our study responds to this question by experimenting with ways to maximize the chance that a receiver sees the incentive and considers it legal tender. Accordingly, how the cash is displayed might influence the potential respondents' decision to open the survey invitation mailing. There are a few ways that prepaid cash incentives can be displayed via a window envelope during mail contact requesting a survey response. As such, we set out to investigate the following hypotheses:

Hypothesis 1: Making the prepaid cash incentive visible through a window envelope will increase recruitment rates compared to a prepaid cash incentive not visible through the envelope. Therefore, making the cash visible will attain a lower cost per recruit.

The use of incentives in surveys is well documented. Prepaid incentives are particularly effective at increasing response rates (Singer and Ye 2013; Mercer et al. 2015). Cash incentives are meant as a gesture of appreciation of the potential respondent, similar to the notion of social exchange, which argues that showing tokens of appreciation prior to a survey request establishes social trust and increases benefits of responding to a survey request, which then positively relates to the decision to participate (Dillman, et al. 2014). Put another way, the prepaid cash incentive can be seen as a thank you in advance (and cover letters often phrase the provision of the prepaid incentive precisely as such). However, incentives cannot be part of a social exchange if the receiver is not aware of them. Making incentives visible thus serves two purposes: to encourage the receiver to open the letter and, then subsequently, to more favorably consider the enclosed survey request. Hence, we first compare the effect of making cash visible via a window on the survey request mailing envelope without including the window on the envelope, replicating past studies.

Hypothesis 2: $5 visible cash incentive will increase recruitment rates compared to $2 visible cash incentive, but the increase in the prepaid cash incentive will also increase cost per recruit.

Higher incentives increase response rates, but the impact of incentives on response rates is curvilinear and shown to decelerate with higher amounts, especially over a prepaid $2 cash incentive (Mercer et al. 2015). Additionally, requesting membership to a survey panel, in which members are asked to complete two to four surveys a month, is a more burdensome request than a request to complete a single survey. Therefore, we would expect the higher prepaid incentive amount ($5 versus $2) to be more effective during a panel recruitment request than a cross-sectional survey response request. It is unclear whether offering higher incentive amounts will in fact increase the cost per recruit. Lastly, the possibility of theft increases when increasing the amount of cash shown through the envelope. However, prior research did not find evidence of theft (DeBell et al. 2020; DeBell 2022). Therefore, we do not expect a notable impact of mail theft in this study as well.

Hypothesis 3: Visible cash windows placed on the front of the envelope will improve recruitment rates compared to windows placed on the back of the envelope. Hence, the front window method will also decrease cost per recruit compared to the back window method.

Typically, most of the useful information (e.g., the sender's name and return address) is presented on the front of an envelope. Thus, mail recipients are perhaps more likely to pay attention to the front of an envelope than the back. Based on leverage-salience theory, a window through which a cash incentive can be seen, displayed in a prominent location on the envelope, will increase the salience of the visible cash (Groves et al. 2000). Additionally, making the incentive amount visible via a window will require custom manufacturing and hand assembly regardless of the window location of the envelope. Accordingly, there will not be any mailing cost differences among the mailings that include front or back envelopes. However, because we expect that the visible cash window placed on the front of the envelope will be more effective than the back of the envelope in terms of recruitment rates, there will be a decrease in cost per recruit for the front window envelope than the back window envelope.

Hypothesis 4: The large visible cash window placed on the envelope will improve recruitment rates compared to the small visible cash window. Therefore, the large window will decrease cost per recruit compared to the small window.

A small window that might only show a small portion of the enclosed cash may lead the recipient to not believe that the cash is truly legal tender. In contrast, a large window placed on the envelope will increase the visibility of the cash shown through the window, possibly increasing the saliency of the visible cash incentive. Making the incentive amount visible via a window requires custom manufacturing and hand assembly regardless of the window size, yielding similar costs when including envelopes with different size windows. However, given that we expect that the large window will be more effective than the small window in bringing in more recruits, this will lead to a decrease in cost per recruit for the large window group compared to the small window group.

Hypothesis 5: Inclusion of two $1 bills (total value of $2) in the envelope and making one of the bills ($1 bill) visible through the envelope will increase recruitment rates compared to inclusion of one $2 bill in the envelope and making $2 bill visible. Accordingly, inclusion of two $1 bills will decrease cost per recruit compared to inclusion of one $2 bill.

Whether a recipient recognizes that the tender being shown is real and legitimate is a notable concern. This concern is exacerbated when the tender is a $2 bill. Indeed, about one in ten Americans are not aware that there is a legal $2 bill (SSRS 2019). We suspect that making $2 visible through the envelope will not be effective for individuals who are not aware of a legal $2 bill. Accordingly, we compared the effect of making $2 versus $1 bills visible via the window on the survey request mailing envelope. However, we have included the same cash value in both conditions (one $2 bill versus two $1 bills) because past research has shown notable marginal gains when comparing $2 versus $1 prepaid cash incentive offered during mailed survey requests (Jobber et al. 2004; Singer and Ye 2013; Mercer et al. 2015). Additionally, we expect that a visible $1 bill will be more likely to bring in more recruits and thus decrease cost per recruit compared to the one $2 bill group.

Hypothesis 6: Making the tender amount visible will improve response rates more than making the face of the tender visible through the window envelope. Thus, making the numeric tender amount visible will decrease cost per recruit compared to presenting the face/image on the tender.

Similarly, respondents might recognize that the tender is real cash but not recognize the amount. Because higher amounts generally foster greater survey participation (Mercer et al. 2015), it is useful for respondents to recognize the amount being offered through the incentive. As such, showing as much of the tender as possible, and minimally, displaying the cash value amount shown at either end of the tender rather than the central face of the bill may help sample members to accurately identify the specific tender amount. Because a window that makes the cash/tender visible requires custom manufacturing and hand assembly, there are no expected mailing cost differences across the mailings that show different parts of the tender. Since we expect that showing the tender's numeric amount through the envelope will more likely bring in more recruits, it will decrease cost per recruit compared to showing the image on the tender.

2. DATA AND METHODS

2.1 Data and Experiment Design

We tested the hypotheses related to the efficacy of amount, placement, and visibility of cash incentives seen through the envelope during the 2021 recruitment cycle of AmeriSpeak, a probability-based panel developed and managed by NORC at the University of Chicago. AmeriSpeak sample is drawn from US households with a known, nonzero probability of selection from the NORC National Frame (NORC 2022). The NORC National Frame is an area probability sampling frame with near complete (97 percent) coverage of all US households. It is based on the US Postal Service Delivery Sequence File (DSF) and uses in-person listing in areas in which the DSF coverage is determined to be insufficient. To maximize recruits from various key populations of interest, AmeriSpeak utilizes a disproportionate stratified sampling design whereby the strata were households with a specific prediction of a given attribute based on NORC Big Data Classifiers (Dutwin et al. 2023). Classifiers predicted the likelihood of a household/person in the household being aged 18–24, aged 50 or older, African-American, Hispanic, a holder of at most a high school diploma, Spanish speaking, or a parent or guardian of a child in the household.

Panel recruitment begins with sending a prenotification postcard to sampled households alerting potential respondents of a future mailing packet. A 9″ × 12″ first class envelope packet is sent 5 days after the prenotification postcard and includes a cover letter, an informative brochure about AmeriSpeak, and typically a \$2 prepaid incentive (those who join are also provided a promised incentive, AmeriPoints worth \$20 + \$5 early bird if they respond within 1 week of letter reception). The recruitment cover letter invites sampled households to join AmeriSpeak panel online by visiting the panel website or by calling a toll-free telephone line (inbound). Additionally, a reminder postcard is sent 1 week after the recruitment package. After the initial mailings, many nonresponding households are contacted via phone, FedEx mailings, and in-person techniques (for more information, see NORC 2022). Any US household members ages 18 and older from the recruited households are eligible to be an AmeriSpeak panelist. AmeriSpeak reports a recruitment rate of 29 percent, calculated via the recruitment rate formula provided in AAPOR standard definitions report (AAPOR RECR, see AAPOR 2016). This study specifically focuses on mail recruitment and experiments conducted during the initial 9″ × 12″ envelope recruitment mailing.

AmeriSpeak recruiting samples are typically split into sample replicates to allow field interviewers to have their labor spread across several months of fielding. We utilized three mailing replicates (MR1, MR2, and MR4) in 2021 for our visible cash experiments: 15,000 sampled addresses in MR1 and MR2, respectively, and 6,000 in MR4. We conducted a distinct experiment when we sent the initial 9″ × 12″ envelope recruitment mailing to each mailing replicate. Accordingly, we will reference experiments that utilized MR1 as *experiment #1*, MR2 as *experiment #2*, and MR4 as *experiment #3* throughout the article. The survey data collection for these three experiments took place from February 18 to July 20, 2021.

2.1.1 Experiment #1

As illustrated in table 1, during the first experiment, sampled households were first randomized to receive either a $2 or a $5 prepaid incentive ($N = 7{,}500$ per condition). Second, sampled households were randomly split into groups that received no window versus a window in the back of the envelope versus a window in the front of the envelope (see appendix 1 in the supplementary data online for envelope and window appearances). Forty percent ($N = 6{,}000$) were assigned the no window condition, with an additional 30 percent assigned to each of the front versus back window conditions ($N = 4{,}500$ each). These two conditions were fully crossed with incentive amount, resulting in six treatment groups (3 × 2) overall. The cell sizes for these fully crossed (i.e., full factorial) treatment groups ranged between 1,125 and 1,500 (on average $N = 1{,}250$ per cell).

2.1.2 Experiment #2

During the second experiment, all sampled households received a front window envelope due to the success of this treatment in experiment #1. The sampled households were randomly assigned to either a small (2″ diameter circle) or large (3″ × 2.3″ rectangle) window on the envelopes (see appendix 1 in the supplementary data online for envelope and window appearances). A random half of the sample ($N = 7{,}541$) received the small window envelope while the other random half received the large window envelope ($N = 7{,}459$). Furthermore, we continued to vary the amount of the prepaid incentive, again as either $2 or $5 ($N =$ approximately 7,500 per condition). The fully crossed experiment conditions yielded four groups (2 × 2) with cell sizes ranging between 3,714 and 3,782 (on average $N = 3{,}750$ per cell).

2.1.3 Experiment #3

During the third experiment, all sample members received a small front window envelope but were randomly assigned first to either a treatment with the face (e.g., Abraham Lincoln) of the tender showing or a treatment with the tender amount exposed (see appendix 1 in the supplementary data online for

Table 1. Experiment Design: Experiment Condition Descriptions and Sample Sizes

Experiment conditions	Sample size
Experiment #1 (hypotheses 1–3)	
Three experiment conditions	Total $N = 15{,}000$
(3 × 2 experimental design)	($N = 2{,}500$ per 6 cells, on average)
• No window (control) versus front window versus back window envelope	• No window: 6,000 (40%)
	• Front window: 4,500 (30%)
• $2 (control) versus $5 prepaid incentive	• Back window: 4,500 (30%)
	• $2 versus $5 prepaid: 7,500 (50%)
Experiment #2 (hypotheses 2 and 4)	
Two experiment conditions	Total $N = 15{,}000$
(2 × 2 experimental design)	($N = 3{,}750$ per 4 cells, on average)
All samples received the front window envelope.	• $2 prepaid: 7,504 (50%)
• Continuation of the $2 (control) versus $5 prepaid incentive	• $5 prepaid: 7,496 (50%)
	• Small window: 7,541 (∼50%)
• Small (control) versus large size front window envelope	• Large window: 7,459 (∼50%)
Experiment #3 (hypotheses 2, 5, and 6)	
Two experiment conditions	Total $N = 6{,}000$
(3 × 2 experimental design)	($N = 1{,}000$ per 6 cells, on average)
All samples received the small front window envelope.	• Two $1: 1,498 (25%)
	• One $2: 1,501 (25%)
• One $2 (control) versus two $1 versus one $5 prepaid incentive	• One $5: 3,001 (50%)
	• Face of tender: 2,998 (50%)
• Tender amount (control) versus face of the tender showing through the front window envelope	• Amount of tender: 3,002 (50%)

envelope and window appearances). In this experiment, roughly half of the sample ($N = 3{,}000$) was assigned to each tender treatment group. Secondly, we continued to vary the amount of the prepaid incentive, again as either $2 or $5 ($N$ = approximately 3,000 per condition). Additionally, in this experiment, we have divided the $2 treatment group into two additional treatment groups and sent two $1 prepaid cash incentive to one random half and one $2 prepaid cash incentive to the other random half (N = approximately 1,500 per condition). The fully crossed experiment conditions yielded to six groups (3 × 2) with cell sizes ranging between 750 and 1,500 depending on the treatment group (on average $N = 1{,}000$ per cell).

2.2 Outcome Measures and Analyses

We cannot know which sample members opened the letter, and as such our primary outcome variable of interest is whether a household responded to the recruitment survey and joined AmeriSpeak. Thus, the principal outcome measure for all experiments in this study is the panel recruitment rate (AAPOR RECR, see AAPOR 2016). Panel recruitment rates (AAPOR RECR) are calculated based on responses from mailings prior to phone contact during the initial recruitment and is referenced as *RECR* throughout the article.

The full factorial design in our study allows us to examine the effect of each experimental condition, as well as the effects of interactions between experimental conditions on the outcome measure. Thus, for each of the three experiments, we first present the overall effect of each of the experimental conditions (i.e., main effect) on the outcome measures. Second, we illustrate the fully crossed effect of experiment conditions (i.e., effect of interactions between factors) on outcome measures. During each step (examination of main effects and fully crossed effects), we compared both outcome measures between the control group (listed in table 1) and each experiment group. During the comparison of the primary outcome measure (RECR), we conducted pairwise comparisons using chi-square tests.

While the principal outcome variable for all experiments is the panel recruitment rate, we also consider the percentage change in the cost per recruit relative to the baseline cost of the control condition as our secondary outcome measure. Cost per recruit calculations include the cost to print, assemble, and mail the recruitment material, as well as prepaid and promised incentive costs and success of recruitment within each experiment condition. Because different experimental treatments may have different effects on costs, we calculate the cost per recruit for each combination of observed experimental treatments (which we refer to as the fully crossed group) using the following cost components:

SU = Number of sampled households sent mailing within each fully crossed group;
P = Printing cost for each mailing;
L = Labor cost to prepare and mail each mailing;
S = Postage (i.e., stamp) cost for each mailing;
PPI = Prepaid incentive cost included in each mailing piece;
R = Number of households recruited into the panel; and
PI = Promised incentive cost given to each household joining the panel

$$\text{Fully crossed group cost per recruit} = \left(\frac{\text{SU} \times (P + L + S + \text{PPI})}{R}\right) + \text{PI}.$$

The cost per recruit calculation for a main effects experimental condition includes the above cost components and aggregates the cost per recruit of each of the fully crossed experimental treatment groups for the corresponding main effect. To calculate the cost per recruit for each main effect group, we first calculate the total costs for each fully crossed experimental treatment group by multiplying the number of households recruited in a specific fully crossed group with the cost per recruit of that specific fully crossed group (fully crossed group cost per recruit). We then add these costs together for all the fully crossed groups within the corresponding main effect group to arrive to the total mailing costs for a specific main effect group. Lastly, we divide this total with the total number of recruited households in the main effect group, summing the number of recruited households within a specific main effect group (for all N fully crossed groups within each main effect group).

$$\text{Main effect group cost per recruit} = \frac{\sum_{1}^{N} (R \times \text{fully crossed group cost per recruit})}{\sum_{1}^{N} R}.$$

Because the actual dollar cost per recruit is considered to be NORC proprietary information, cost comparisons among fully crossed or main effect groups are calculated by measuring the percentage difference in cost per recruit relative to the corresponding control group [percentage change = (experiment grp cost − control grp cost)/control grp cost]. To take into account the complex sample design components in the AmeriSpeak sample, we used PROC SURVEYFREQ procedure in SAS 9.4 and defined cluster (primary sampling unit variable), strata (stratification variable), and the base weight in the analyses. AmeriSpeak recruitment uses a stratified random sample of housing units selected from the NORC National Frame and groups that are less likely to respond are sampled at a higher rate (i.e., oversampled) to increase representation of the panel. The base weighted results are adjusting for this oversampling factor and are calculated as the inverse of probability of selection of the housing units.

3. RESULTS

3.1 Experiment #1

We begin with a test of the main effects of three experimental conditions implemented in the first experiment. Table 2 provides overall main effects of inclusion of a window on the envelope, as well as the different amounts of prepaid incentives. Within the windowed conditions, the front window envelope had the highest recruitment rate (AAPOR RECR) of 5.7 percent and had a 20.5 percent (=0.97 pp/4.73 percent) improvement over the no window (control) envelope. In other words, the odds of a recruitment mailing sent using a front window envelope with visible cash incentive were 1.2 times and significantly higher to convert a sample household member to be an AmeriSpeak panelist compared to a mailing using a windowless envelope ($p \leq .05$). We did not observe a notable difference in recruitment rates based on the location of

Table 2. Window and Incentive Results (Main Effects)

Experiment group	Sample size	RECR (%)	Treatment versus control				
			RECR				Cost per recruit (%)
			DIFF (pp)	Odds ratio	Chi-square	*p*-Value	
No window (control)	6,000	4.73	N/A	N/A	N/A	N/A	N/A
Front window	4,500	5.70	0.97	1.22	3.809	.051	−12.04
Back window	4,500	5.30	0.57	1.13	1.376	.241	−11.62
\$2 prepaid (control)	7,500	4.29	N/A	N/A	N/A	N/A	N/A
\$5 prepaid	7,500	6.08	1.79	1.44	18.978	<.0001	11.65

NOTE.—N/A: not applicable.

the window on the envelope (front versus back). Additionally, a recruitment mailing with visible cash through a back window on the envelope was 1.1 times more likely to convert a sample household member to an AmeriSpeak panelist than a mailing using windowless envelope. The odds of being recruited were significantly higher for the recruitment mailing envelope that makes cash visible using the front window than a recruitment request mailing envelope with no window (control group). However, this was not the case for back window condition when compared to the no window group, largely a product of lack of statistical power caused by small number of recruits. Additionally, it was 12.0 percent less costly to recruit a panelist via the front window envelope and 11.6 percent less costly to recruit via the back window envelope than the no window control group.

The \$5 cash prepaid incentive improved the recruitment rate compared to a \$2 cash prepaid incentive. Overall, a recruitment mailing sent with a \$5 prepaid incentive was 1.4 times more likely to convert a sample household member to an AmeriSpeak panelist than the \$2 prepaid incentive ($p < .05$). It was 11.7 percent more costly to obtain a recruit with the \$5 prepaid incentive than the \$2 prepaid incentive. Given there appears to be a clear pattern of response by each of the principal conditions (i.e., main effects), we further tested fully crossed effects of experiment conditions, as shown in table 3.

Overall, the condition with a front window and the highest prepaid (\$5) incentive attained the highest recruitment rate, while the condition without a window and the lowest prepaid (\$2) incentive attained the lowest recruitment rate. Notably, the odds of being recruited were significantly higher (nearly twice as much) for the top treatment group, the \$5 front window condition,

Table 3. Window/Incentive Results Across All Conditions (Fully Crossed 3 × 2 Design)

Experiment group	Sample size	RECR (%)	Treatment versus control				
			RECR				Cost per recruit (%)
			DIFF (pp)	Odds ratio	Chi-square	*p*-Value	
\$2—no window (control)	3,000	3.51	N/A	N/A	N/A	N/A	N/A
\$2—front window	2,250	4.87	1.36	1.41	4.603	.032	−15.24
\$2—back window	2,250	4.77	1.26	1.38	4.190	.041	−22.22
\$5—no window	3,000	5.95	2.44	1.74	15.452	<.0001	1.70
\$5—front window	2,250	6.53	3.02	1.92	20.191	<.0001	−8.32
\$5—back window	2,250	5.81	2.30	1.70	12.568	.000	−1.20

NOTE.—N/A: not applicable.

than the control group with the lowest recruitment rate, the \$2 no window condition ($p < .05$). Additionally, it was 8.3 percent less costly to obtain a recruit with the \$5 front window condition than the control group. Notably, the top performing three conditions (i.e., with highest RECR) all offered the \$5 prepaid incentive.

3.2 Experiment #2

In the second experiment, we adapted our mailing strategy and experiments based on the results from the first experiment. Accordingly, all sample members received the front window envelope in the second experiment. We continued to test the efficacy of prepaid incentives, but this time using a front window and varying the size of the window on the recruitment envelope.

Table 4 provides overall recruitment for the prepaid incentive and window envelope groups in experiment 2. Similar to the first experiment results, the \$5 prepaid incentive group improved the recruitment rate compared to the \$2 prepaid incentive group. Overall, a recruitment mailing sent with a \$5 prepaid incentive was 1.4 times more likely to convert a sample household into an AmeriSpeak panelist in experiment #2 ($p < .05$). It was also roughly 17.7 percent more costly to recruit a sample member with a \$5 prepaid incentive than a \$2 prepaid incentive in this round. The odds of being recruited were not significantly different between small and large envelope windows used to demonstrate the cash included in the envelope ($p > .05$), though large window group

Table 4. Window Size/Incentive Results (Main Effects)

Experiment group	Sample size	RECR (%)	Treatment versus control				
			RECR				Cost per recruit (%)
			DIFF (pp)	Odds ratio	Chi-square	*p*-Value	
$2 prepaid (control)	7,504	3.28	N/A	N/A	N/A	N/A	N/A
$5 prepaid	7,496	4.47	1.19	1.38	6.295	.012	17.67
Small front window (control)	7,541	3.92	N/A	N/A	N/A	N/A	N/A
Large front window	7,459	3.83	−0.08	0.98	0.040	.841	3.55

NOTE.—N/A: not applicable.

cost per recruit was slightly higher mainly because the small front window had a slightly higher recruitment rate.

Table 5 provides recruitment rate comparisons by incentive amount and window envelope size for the second experiment. As shown in table 5, the small front window with a $5 prepaid incentive attained the highest recruitment rate across all groups. Compared to the small front window with $2 (control) group, the odds of being recruited were significantly higher for both the large and small front window with $5 prepaid incentive groups ($p < .05$), while the odds of being recruited were slightly but not significantly higher for the large front window with $2 treatment group. While the small front window with $5 condition attained the highest recruitment rate among four experiment groups, it was also 7.4 percent more costly to administer than the control (small front window with $2).

3.3 Experiment #3

In the last experiment, we tested what part of the tender was shown through the window (cash amount versus face/image on the cash) for different prepaid cash amounts ($5 versus $2). We then split the $2 prepaid treatment by whether the letter contained one $2 bill versus two $1 bills given that some sample members may not be familiar with the $2 bill and may think it is fake. Hence, we also tested showing $1 through the window. As only one bill can be shown, recipients would likely assume the letter only contains $1, but upon successful opening would find $2 in total.

Table 6 provides overall recruitment rates for the various prepaid incentives and different tender reveal groups. Similar to the earlier experiments, the inclusion of a $5 prepaid cash incentive shown through the window on the

Table 5. Window Size/Incentive Results Across All Conditions (Fully Crossed 2 × 2 Design)

Experiment group[a]	Sample size	RECR (%)	Treatment versus control				
			RECR				Cost per recruit (%)
			DIFF (pp)	Odds ratio	Chi-square	p-Value	
Small window w/ $2 (control)	3,759	2.99	N/A	N/A	N/A	N/A	N/A
Small window w/ $5	3,782	4.85	1.87	1.66	11.830	.001	7.42
Large window w/$2	3,745	3.58	0.60	1.21	2.385	.123	−6.87
Large window w/$5	3,714	4.09	1.10	1.39	3.968	.046	20.51

NOTE.—N/A: not applicable.
[a]Both small and large windows that display cash are located at the front of the envelope.

Table 6. Tender Reveal/Incentive Results (Main Effects)

Experiment group	Sample size	RECR (%)	Treatment versus control				
			RECR				Cost per recruit (%)
			DIFF (pp)	Odds ratio	Chi-square	p-Value	
One $2 prepaid (control)	1,501	2.37	N/A	N/A	N/A	N/A	N/A
Two $1 prepaid	1,498	3.29	0.92	1.40	1.447	.229	−23.37
One $5 prepaid	3,001	4.50	2.14	1.94	8.294	.004	−8.71
Number visible (control)	3,002	4.25	N/A	N/A	N/A	N/A	N/A
Face/image visible	2,998	3.07	−1.19	0.71	4.499	.034	31.28

NOTE.—N/A: not applicable.

recruitment envelopes produced the highest recruitment rate than other $2 prepaid incentive groups (one $2 bill and two $1 bills) shown through the window. Overall, a mailing sent with a $5 prepaid incentive was 1.94 times more likely to convert a sample household member to an AmeriSpeak panelist than a mailing sent with one $2 prepaid incentive ($p < .05$). On the other hand, even

Table 7. Tender Reveal/Incentive Results Across All Conditions (Fully Crossed 3 × 2 Design)

Experiment group	Sample size	RECR (%)	Treatment versus control				
			RECR				Cost per recruit (%)
			DIFF (pp)	Odds ratio	Chi-square	*p*-Value	
One \$2—number visible (control)	752	2.48	N/A	N/A	N/A	N/A	N/A
One \$2—face/image visible	749	2.25	−0.23	0.91	0.078	.781	−17.72
Two \$1—number visible	749	3.66	1.18	1.49	1.018	.313	−42.42
Two \$1—face/image visible	749	2.91	0.43	1.18	0.204	.652	−12.08
One \$5—number visible	1,501	5.45	2.96	2.26	7.779	.005	−29.48
One \$5—face/image visible	1,500	3.56	1.08	1.45	1.322	.250	−0.39

NOTE.—N/A: not applicable.

though cost per complete savings were notable, the two \$1 prepaid cash incentive shown through the window did not produce an overall significant increase in the recruitment rates compared to the one \$2 prepaid incentive. Overall, the treatment group that received the visible cash amount (i.e., number) had a higher recruitment rate than the group which was shown the face/image on the cash showing through the window on the envelope. Specifically, the odds of being recruited were significantly lower for a mailing sent with the face/image on the cash showing through the window envelope than a mailing sent with the tender's numeric amount showing through the window envelope (odds ratio = 0.71; $p < .05$).

Table 7 provides recruitment rate comparisons by incentive amount and tender reveal for the last experiment. Based on the results illustrated on this table, showing a \$5 prepaid incentive, and specifically showing its tender amount through the window, attained a higher recruitment rate than any other treatment group. This condition was also among one of the more cost-effective conditions to administer. Specifically, showing the tender amount of the \$5 prepaid incentive was 29.4 percent less costly to administer than the control (showing the tender amount of the \$2 prepaid incentive). Additionally, the two \$1 prepaid cash incentive, and specifically showing the \$1 tender amount through the window, also attained the second highest recruitment rate among the treatment groups in this replicate and this treatment group was the most cost efficient among all of the groups (42.4 percent less costly to administer than the control). Even when the tender amount was visible (i.e., numeric amount is made visible through the window), the two \$1 prepaid cash incentive group attained a higher recruitment rate than the one \$2 prepaid incentive group; however, this difference was not statistically significant. These null findings are likely due in part to small sample sizes, so we believe that this option (two \$1, numeric amount is made visible through the window) is still a good alternative for lower budget projects.

Lastly, we compared key demographic distributions between the groups defined by the main treatment effects to assess whether there are differences among recruited individuals per each mailing intervention tested for each experiment (see appendix 2 in the supplementary data online). Specifically, we examined whether mailing interventions are able to bring in more individuals that are harder to reach, recruit, and retain (such as young adults, minorities, lower income individuals, and persons with lower educational attainment). We did not observe an overall systematic and notable impact of the mailing interventions on demographic composition, specifically among hard-to-reach groups in the experiment #1 sample. When we examined the experiment #2 recruits, we observed that the higher \$5 prepaid incentives brought in slightly more recruits who were younger, African-American/Black, and Asian and Pacific Islander but also slightly fewer recruits with lower income and lower educational attainment compared to the \$2 prepaid incentive group. Additionally, the large window brought in slightly more recruits who were young adults and lower-income individuals than the small window.

Higher \$5 prepaid incentives also brought in more recruits who were young adults, Hispanic, Asian and Pacific Islander and persons with lower educational attainment compared to the \$2 prepaid incentive groups in the experiment #3 sample. Two \$1 prepaid cash incentives performed worse overall than the one \$2 bill prepaid incentive among hard-to-reach groups (such as young adults, African-Americans/Blacks, Hispanics, and lower-income individuals), except persons with lower educational attainment in the experiment #3 sample. Overall, the experimental group receiving the mailing with the visible tender amount performed slightly better among young adults, Hispanics, and African-Americans/Blacks but performed worse among persons with lower income than the group receiving the mailing with the visible face of the tender. Due to the low RECR, the cell sizes per demographic distributions of the recruited panelists are small and are not sufficient to reliably determine whether different experiment groups systematically recruit a different set of panel recruits, specifically for the full factorial experimental conditions. Accordingly, we urge the readers to interpret these results with caution as there is a need for the replication of results with a larger sample.

4. CONCLUSION AND DISCUSSION

Our overall finding is that the recruitment rate improvement from the use of windows that display prepaid cash is robust on a number of levels. First, a household sent a recruitment mailing using a front window envelope with visible cash incentive was 1.2 times more likely to respond than a household receiving a mailing using windowless envelope. Overall, an approximately 20 percent improvement in odds of participation is also a significant reduction in costs to a given research project.

Moreover, various experimental studies that have explored the visible cash mailing approach monitored for mail theft through different mechanisms including undeliverable rates and mail tracking services such as AccuTrace found no significant evidence of theft (DeBell et al. 2020; Sherr and Wells 2021; DeBell 2022; Williams et al. 2022). Similar to the earlier studies, we were also cognizant of the risk of mail theft and monitored for mail theft via reviewing "track and trace" reports from the United States Postal Service, incorporating seed cases within the sample which included staff's mailing addresses, and monitoring undeliverable rates. Notably, we also did not find any significant evidence of mail theft based on our review of these indicators.

Most of the prior research on prepaid cash incentives made visible through the envelope found a significant improvement in response rates with windows compared to no windows (e.g., DeBell et al. 2020; Sherr and Wells 2021; Williams et al. 2022). We extended this research beyond the use of a window versus no window and investigated how this new mailing approach can be strategically optimized even further. In our tests, we explored further variations by cash incentive amount, location of the window on the envelope, window size, and what part of the tender/cash is shown. Notably, we found window placement (front or back of the envelope) or window size to bear no impact on recruitment rate. That said, incentive amount matters, as does showing the tender/cash amount as compared to the image/face on the cash.

Overall, based on the first experiment results, a recruitment mailing sent with a $5 prepaid cash incentive was 1.4 times more likely to increase recruitment as compared to a $2 prepaid cash incentive. We observed the same result in the second experiment as well. Using the small front window strategy in the third experiment, we found that we could improve response 1.3 times more by showing the tender amount/number compared to the face/image on the cash through the window. Related to the visible incentive amount, we found that two $1 bills slightly improved the recruitment rate in comparison to a $2 bill, but this improvement was not statistically significant. Nevertheless, it may be efficient for low budget studies, which cannot afford providing $5 prepaid cash incentives, to reveal $1 rather than $2 cash incentive through the window envelope. Hence, this experiment condition should be replicated and examined further in future studies with larger sample sizes.

Furthermore, as discussed in the article, there are many factors influencing the cost of survey request mailings to sample units including sample size, paper weight, mail dimension, assembly specifications, and postage specifications. AmeriSpeak prefers to use heavier paper, which requires custom manufacturing. Therefore, the cost difference in printing window versus no window envelopes is small, about 1.4 percent. There is no difference in printing cost between the front window and back window envelopes as they both require custom manufacturing. Furthermore, AmeriSpeak specifications for the no window treatment resulted in 100 percent hand assembly. Hence, we do not experience a cost difference in mailing preparation labor between envelopes with no window, front windows, or back windows. There is also no difference in printing envelopes with small windows versus large windows. As with other studies, the cost difference for these interventions may be more than the custom mailings.

Lastly, there are several limitations of this study. First, we note that with such calculations, we ignore possible variations in the overall recruitment rates by experiment sample (i.e., sample replicate). We do so because we witnessed general declines in recruitment rates across multiple studies overall during the time of the 2021 AmeriSpeak recruitment. It has generally been interpreted, at least as one possible reason, as the experiencing of a "COVID bounce" to response rates throughout the latter half of 2020 and the first quarter of 2021. But from the second quarter of 2021 onward, response rates appeared to have regressed to the pre-COVID mean levels (if not continuing their general decades-long trend of decline). The AmeriSpeak recruitment rates (not including the NRFU stage, which substantially boosts recruitment rates) for experiment #1, mailed on February 18, 2021, was 6.1 percent on average for the $5 front window conditions. Experiment #2, mailed on June 10, 2021, attained 4.5 percent similarly with $5 windows. Experiment #3 (mailed July 1, 2021), also fully windowed at $5, attained a 4.5 percent recruitment rate.

Additionally, and more importantly, while the main effects were well powered, we lacked statistical power for the examination of the interaction effects due to the low recruitment rates. We also did not have sufficient cell sizes to provide robust subgroup analyses to determine which treatments did better at bringing in the harder-to-reach groups. The subgroup analyses results can help assess optimal strategies to recruit hard-to-reach populations and provide an opportunity to tailor recruitment materials to varying population subgroups and/or hard-to-reach groups. Hence, there is an opportunity for future research in terms of replicating these experiments and analyses with larger samples. We also recommend potential replications of these experiments using cross-sectional surveys. Given the less demanding nature of the single survey completion request compared to the panel recruitment request, we expect cross-sectional studies to achieve higher response rates with similar sample sizes than our study. Nevertheless, we believe that the results from this study will shed light on future recruitment mailing strategies as well as potential replication of similar experiments in future studies.

Supplementary Materials

Supplementary materials are available online at academic.oup.com/jssam.

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